

Seed robustness of the controlled ($O(2)$ -twiddle) DCT-IV basis

100 random real-orthogonal initialisations, normal top-20% training (1008 steps) on the full 500-image DIV2K-8q train pool

Generated 2026-07-01

Setup

The **controlled** DCT-IV parametrization (`DCT4Basis(8, 8, parametrization: "controlled")`) trains the affine twiddle as a single-angle CRY gate on $O(2)$ and keeps the mirror Q/R CNOTs as fixed CX routing — **408 trainable parameters vs 2872** for the dense- $O(4)$ form ($7\times$ fewer), and $\sim 2\times$ faster per optimizer step. As in the o4 seed study, each of 100 seeds s reseeds **everything trainable** (a fresh Haar real-orthogonal init via `dct4_random_controlled_basis(8, 8, s)`, the training RNG `train_basis_batched(seed = s)`), but here trained on the **full 500-image** pool with mini-batches of 50 (not a 50-image subsample), one cosine-LR top-20% run of 1008 steps. The held-out **test set is held fixed** (canonical seed 42, 50 images). Test PSNR is scored at four keep ratios ρ .

Endpoint variance

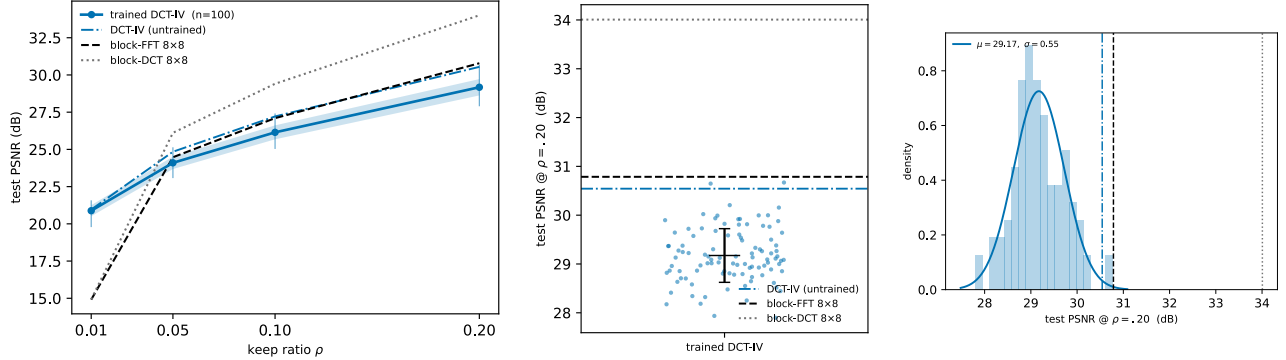


Figure 1: Test PSNR across 100 seeds. (a) mean $\pm\sigma$ band (+ min–max whiskers) vs ρ , against the untrained exact DCT-IV (dashdot), block-FFT 8×8 (dashed) and block-DCT 8×8 (dotted). (b) per-seed scatter at $\rho\{=\}.20$ with the mean $\pm\sigma$ bar. (c) the endpoint distribution at $\rho\{=\}.20$ (histogram + fitted normal); references marked.

basis	$\rho\{=\}.01$	$\rho\{=\}.05$	$\rho\{=\}.10$	$\rho\{=\}.20$	min@.20
controlled DCT-IV	20.88 $\pm$ 0.37	24.1 $\pm$ 0.42	26.15 $\pm$ 0.47	29.17 $\pm$ 0.55	27.9
DCT-IV (untrained)	20.96	24.84	27.21	30.54	—
block-FFT 8×8	14.91	24.47	27.1	30.79	—
block-DCT 8×8	14.93	26.11	29.41	34.01	—

mean $\pm\sigma$ test PSNR (dB), $n = 100$ random-init seeds; **min@.20** is the single worst seed at $\rho\{=\}.20$.

Reading

From 100 Haar real-orthogonal inits, 1008 steps of top-20% training on the **full 500-image** pool concentrate the endpoint into a narrow band: at $\rho\{=\}.20$ the per-seed standard deviation is 0.55 dB (mean 29.17, min 27.9, max 30.67). Against the untrained exact DCT-IV (30.54 dB @ $\rho\{=\}.20$), 2 of 100 trained seeds match-or-exceed it; against block-FFT 8×8 (30.79 dB), 0 clear it.

Two things this study establishes. (i) From a random start the controlled DCT-IV plateaus **just below** the fixed transform — reaching the 31–33 dB of exact-init training requires the **canonical exact-DCT-IV init**, not more data (moving the pool from 50 to 500 images added only ~ 0.04 dB @ $\rho\{=\}.20$; the lever is initialisation, not data). (ii) Despite fixing the mirror routing and using **$7\times$ fewer** trainable parameters, the controlled basis still **beats the dense- $O(4)$ form from random init** at every ρ (+ 0.4–1 dB) with lower variance — the fixed CX mirror is a correct structural prior that leaves less of the DCT-IV to rediscover.

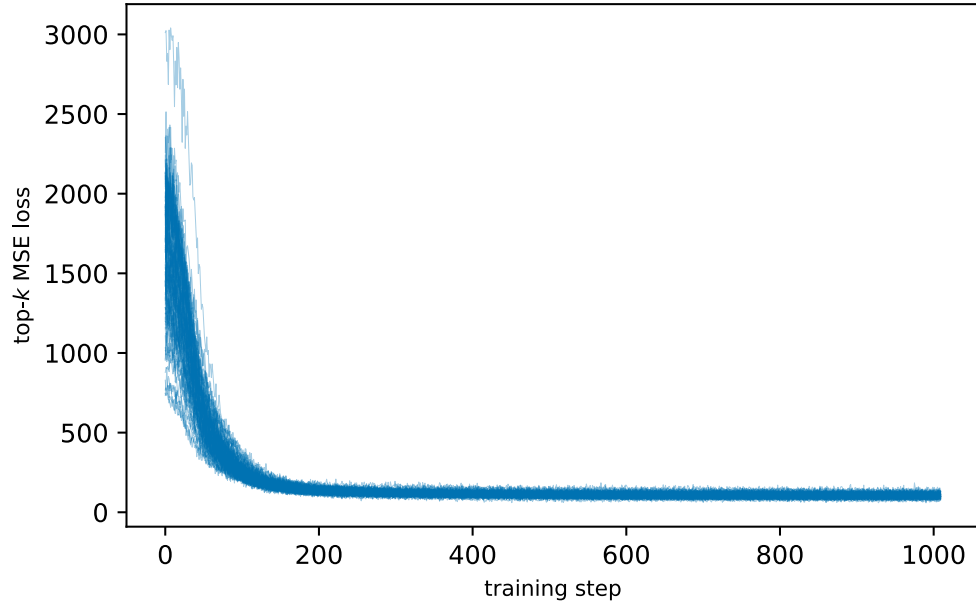


Figure 2: Per-seed training dynamics: every seed’s top- k MSE descent (one curve per seed) on a shared y-range. The Haar real-orthogonal starts descend from genuinely different, far-from-trained losses and converge into the narrow endpoint band above.